THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 6	Common prime factors give HCF $= 6$
2	**(b)** 12	LCM $= \frac{4 \times 6}{2} = 12$
3	**(b)** 3	$12 = 2^2 \times 3$
4	**(d)** $\sqrt{7}$	7 is not a perfect square, so $\sqrt{7}$ is irrational
5	**(c)** $2^2 \times 3 \times 5$	Divide repeatedly by primes: $60 = 2^2 \times 3 \times 5$
6	**(c)** terminating	Denominator 8: only 2 and 5 as prime factors
7	**(b)** 10	Common prime factors give HCF $= 10$
8	**(b)** 18	LCM $= \frac{6 \times 9}{3} = 18$
9	**(b)** 3	$30 = 2 \times 3 \times 5$
10	**(b)** $\sqrt{11}$	11 is not a perfect square, so $\sqrt{11}$ is irrational

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$18 = 2 \times 3^2$, $24 = 2^3 \times 3$ (1); HCF $= 6$ (1).	1+1
12	Divide repeatedly by primes (1); $140 = 2^2 \times 5 \times 7$ (1).	1+1
13	$16 = 2^4$ (1); it has only 2 and/or 5, so terminating (1).	1+1
14	$30 = 2 \times 3 \times 5$, $45 = 3^2 \times 5$ (1); HCF $= 15$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$26 = 2 \times 13$, $91 = 7 \times 13$ (1); HCF $= 13$ (1); LCM $= 182$ (1).	1+1+1
16	HCF $= 6$, LCM $= 60$ (1+1); HCF$\times$LCM $= 360 = 12 \times 30$ (1).	1+1+1
17	$16 = 2^4$ (1); terminating (1); value ≈ 0.1875 (1).	1+1+1
	OR: $96 = 2^5 \times 3$, $404 = 2^2 \times 101$ (1); HCF $= 4$ (1); LCM $= 9696$ (1).	1+1+1
18	HCF $= 7$, LCM $= 70$ (1+1); HCF$\times$LCM $= 490 = 14 \times 35$ (1).	1+1+1
	OR: $25 = 5^2$ (1); terminating (1); value ≈ 0.28 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$135 = 3^3 \times 5$, $225 = 3^2 \times 5^2$ (2); HCF $= 45$ (1); LCM $= 675$ (1); $45 \times 675 = 30375$ verified (1).	2+1+1+1
	OR: $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, $24 = 2^3 \times 3$ (3); HCF $= 6$ (1); LCM $= 72$ (1).	3+1+1
20	$196 = 2^2 \times 7^2$, $280 = 2^3 \times 5 \times 7$ (2); HCF $= 28$ (1); LCM $= 1960$ (1); $28 \times 1960 = 54880$ verified (1).	2+1+1+1
	OR: $20 = 2^2 \times 5$, $30 = 2 \times 3 \times 5$, $45 = 3^2 \times 5$ (3); HCF $= 5$ (1); LCM $= 180$ (1).	3+1+1

Mistakes to avoid:

- Using HCF $\times$ LCM = product for **three** numbers; the result holds only for two numbers.
- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.
- Checking the denominator before reducing the fraction to lowest terms when testing for a terminating decimal.
- Assuming the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{2} = 2$.
- In irrationality proofs, forgetting to state that a and b are co-prime at the start, which is what gives the contradiction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $2^2 \times 3 \times 7$	Divide repeatedly by primes: $84 = 2^2 \times 3 \times 7$
2	**(a)** non-terminating repeating	Denominator 15: has a prime factor other than 2 and 5
3	**(c)** 12	Common prime factors give HCF $= 12$
4	**(d)** 24	LCM $= \frac{8 \times 12}{4} = 24$
5	**(b)** 5	$48 = 2^4 \times 3$
6	**(a)** $\sqrt{13}$	13 is not a perfect square, so $\sqrt{13}$ is irrational
7	**(b)** $2 \times 3^2 \times 5$	Divide repeatedly by primes: $90 = 2 \times 3^2 \times 5$
8	**(c)** terminating	Denominator 20: only 2 and 5 as prime factors
9	**(a)** 5	Common prime factors give HCF $= 5$
10	**(d)** 30	LCM $= \frac{10 \times 15}{5} = 30$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Divide repeatedly by primes (1); $252 = 2^2 \times 3^2 \times 7$ (1).	1+1
12	$30 = 2 \times 3 \times 5$ (1); it has another prime factor, so non-terminating repeating (1).	1+1
13	$36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$ (1); HCF $= 12$ (1).	1+1
14	Divide repeatedly by primes (1); $396 = 2^2 \times 3^2 \times 11$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$72 = 2^3 \times 3^2$, $108 = 2^2 \times 3^3$ (1); HCF $= 36$ (1); LCM $= 216$ (1).	1+1+1
16	HCF $= 6$, LCM $= 144$ (1+1); HCF×LCM $= 864 = 18 \times 48$ (1).	1+1+1
17	$40 = 2^3 \times 5$ (1); terminating (1); value ≈ 0.225 (1).	1+1+1
	OR: $60 = 2^2 \times 3 \times 5$, $84 = 2^2 \times 3 \times 7$ (1); HCF $= 12$ (1); LCM $= 420$ (1).	1+1+1
18	HCF $= 10$, LCM $= 100$ (1+1); HCF×LCM $= 1000 = 20 \times 50$ (1).	1+1+1
	OR: $3 = 3$ (1); non-terminating repeating (1); value ≈ 0.6667 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$867 = 3 \times 17^2$, $255 = 3 \times 5 \times 17$ (2); HCF $= 51$ (1); LCM $= 4335$ (1); $51 \times 4335 = 221085$ verified (1).	2+1+1+1
	OR: $16 = 2^4$, $40 = 2^3 \times 5$, $56 = 2^3 \times 7$ (3); HCF $= 8$ (1); LCM $= 560$ (1).	3+1+1
20	$240 = 2^4 \times 3 \times 5$, $360 = 2^3 \times 3^2 \times 5$ (2); HCF $= 120$ (1); LCM $= 720$ (1); $120 \times 720 = 86400$ verified (1).	2+1+1+1
	OR: $15 = 3 \times 5$, $21 = 3 \times 7$, $35 = 5 \times 7$ (3); HCF $= 1$ (1); LCM $= 105$ (1).	3+1+1

Mistakes to avoid:

- Using HCF × LCM = product for **three** numbers; the result holds only for two numbers.
- Taking the *greatest* power for HCF or the *smallest* power for LCM; HCF uses smallest common powers, LCM uses greatest powers.
- Checking the denominator before reducing the fraction to lowest terms when testing for a terminating decimal.
- Assuming the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{2} = 2$.
- In irrationality proofs, forgetting to state that a and b are co-prime at the start, which is what gives the contradiction.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Real Numbers** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 5	$72 = 2^3 \times 3^2$
2	**(b)** $\sqrt{15}$	15 is not a perfect square, so $\sqrt{15}$ is irrational
3	**(b)** $2 \times 3^2 \times 7$	Divide repeatedly by primes: $126 = 2 \times 3^2 \times 7$
4	**(b)** non-terminating repeating	Denominator 12: has a prime factor other than 2 and 5
5	**(a)** 14	Common prime factors give HCF $= 14$
6	**(c)** 48	LCM $= \frac{12 \times 16}{4} = 48$
7	**(c)** 4	$90 = 2 \times 3^2 \times 5$
8	**(c)** $\sqrt{17}$	17 is not a perfect square, so $\sqrt{17}$ is irrational
9	**(d)** 8	Common prime factors give HCF $= 8$
10	**(a)** 42	LCM $= \frac{14 \times 21}{7} = 42$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$125 = 5^3$ (1); it has only 2 and/or 5, so terminating (1).	1+1
12	$42 = 2 \times 3 \times 7$, $70 = 2 \times 5 \times 7$ (1); HCF $= 14$ (1).	1+1
13	Divide repeatedly by primes (1); $504 = 2^3 \times 3^2 \times 7$ (1).	1+1
14	$21 = 3 \times 7$ (1); it has another prime factor, so non-terminating repeating (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$45 = 3^2 \times 5$, $75 = 3 \times 5^2$ (1); HCF $= 15$ (1); LCM $= 225$ (1).	1+1+1
16	HCF $= 8$, LCM $= 120$ (1+1); HCF×LCM $= 960 = 24 \times 40$ (1).	1+1+1
17	$6 = 2 \times 3$ (1); non-terminating repeating (1); value ≈ 0.8333 (1).	1+1+1
	OR: $54 = 2 \times 3^3$, $90 = 2 \times 3^2 \times 5$ (1); HCF $= 18$ (1); LCM $= 270$ (1).	1+1+1
18	HCF $= 14$, LCM $= 140$ (1+1); HCF×LCM $= 1960 = 28 \times 70$ (1).	1+1+1
	OR: $9 = 3^2$ (1); non-terminating repeating (1); value ≈ 0.4444 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$306 = 2 \times 3^2 \times 17$, $408 = 2^3 \times 3 \times 17$ (2); HCF $= 102$ (1); LCM $= 1224$ (1); $102 \times 1224 = 124848$ verified (1).	2+1+1+1
	OR: $18 = 2 \times 3^2$, $27 = 3^3$, $36 = 2^2 \times 3^2$ (3); HCF $= 9$ (1); LCM $= 108$ (1).	3+1+1
20	$420 = 2^2 \times 3 \times 5 \times 7$, $588 = 2^2 \times 3 \times 7^2$ (2); HCF $= 84$ (1); LCM $= 2940$ (1); $84 \times 2940 = 246960$ verified (1).	2+1+1+1
	OR: $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$ (3); HCF $= 12$ (1); LCM $= 360$ (1).	3+1+1

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